

Python Programming Assignment - 6

Q1) what is a user-defined function in python? why do we need functions instead of writing code repeatedly?

→ A user-defined function is reusable block of code that you write yourself using the `def` keyword to perform a task.

Without a function, code will repeat 'N' numbers of times, it is not reusable, not maintainable, not readable and major drawback is one mistake in code behaves inconsistently.

Q2) Explain the syntax of a function's definition using the `def` keyword. why is the colon(:) mandatory?

→

```
def function_Name (Parameter1, Parameter2): # header
    # Function body
    return result
```

In python function definitions will start 'def' keyword. after `def` need to define unique function name in the parentheses () it is ~~name~~ placeholder that accepts the values and it is optional.

Colon: marks the end of the function header and requires an indented block next.

(3) what is the importance of Indentation inside a function body? what happens if indentation is incorrect?

→ In python code Indentation is not only for readability. It is a strict syntactic requirement. Indentation tells which line belongs to inside the function. & indented lines will run when the function is called.

If the indentation is wrong python will refuse to run the code and throw the trigger specific error. example: ~~Now~~ Indentation Error.

(4) Differentiate between calling a function and defining a function with a simple explanation.

→ Defining a function means writing the instruction for a task

calling a function means execute the instruction previously defined.

example:

```
def make_Tea(): → defining function  
    print("Brewing Tea")
```

```
make_Tea() → calling function
```

(5) Can python function exist without any statements inside its body? If yes, how?

→ Yes, Python function can exist without any statements inside its body.

No it will not exit the function because ⁱⁿ python after creating a function with unique name the end of line we give colon and colon required an indented block next. if nothing inside a body then it will throw indented error.

(6) what is the roll of Pass statement in a function definition?

→ When function definition executes nothing happens. It acts a temporary placeholder. The pass statement is a null operation.

(7) what is difference between parameters and arguments in Python functions?

→ A ^(Placeholder) parameter is the variable written in a function.

```
def Parameters(a, b): # a and b are parameters.
```

An argument is a actual value passed when calling the function.

```
result = Add(25, 50) # 25 & 50 are arguments.
```

<8> what happens if the numbers of arguments passed does not match the parameters.

→ if does not match then it will throw the error because in positional argument need correct order.

<9> what is return value in function.

→ Return sends value back to the caller.

<10> Can a Python function return multiple values? if yes how does Python handle it internally?

→ yes Python function return multiple value but python handles this packing the values into single tuple

```
def get_user_data():
```

```
    name = "Python"
```

```
    age = 25
```

```
    return name, age # Returning multiple values
```

```
result = get_user_data()
```

```
print(result)
```

Outputs:

(~~name~~ "Python", 25) → A tuple.